

Wireshark

1) Exercise One

Open “Wireshark”, then use the “File” menu and the “Open” command to open the file “WS-Ex- 01.pcap”. You should see 26 packets listed.

This set of packets describes a ‘conversation’ between a user’s client and a central server. This entire conversation happens automatically, after a user types something and hits enter. Look at the packets to answer the following questions in relation to this conversation.

In answering the following questions, use brief descriptions. For example, “In frame X, the client requests a web page, and in frame Y, the server delivers the content of the page.”

- a) **What is the IP address of the client that initiates the conversation?**

131.247.95.216

- b) **Use the first two packets to identify the server that is going to be contacted. List the common name, and three IP addresses that can be used for the server.**

Common name: www.l.google.com
IPs: 64.233.161.99, 64.233.161.104,
64.233.161.147

- c) **What is happening in frames 3, 4, and 5?**

frame 3: solicita un SYN (conexión) al servidor.
frame 4: confirma la conexión (SYN, ACK) y los parámetros.
frame 5: el cliente finaliza el handshake (ACK).

d) What is happening in frames 6 and 7?

frame 6: el cliente realizado una solicitud de tipo HTTP y método GET.

frame 7: confirma que recibió la solicitud(ACK).

e) Ignore frame eight. However, for your information, frame eight is used to manage flow control.

f) What is happening in frames nine and ten? How are these two frames related?

frame 9: comienza la transmisión de datos (respuesta TCP) y continúa en el frame 10, utilizando PDU (Protocolaste Data Unit).

frame 10: muestra la respuesta OK de HTTP.

g) What happens in packet 11?

packet 11: confirma que recibió la respuesta (ACK) del frame 10.

h) After the initial set of packets is received, the client sends out a new request in packet 12. This occurs automatically without any action by the user. Why does this occur? See the first “hint” to the left.

Esto ocurre debido a que luego de procesar el HTML que recibió el cliente y se confirmó en el frame 11, automáticamente se toma el recurso, la imagen “logo.gif”.

i) What is occurring in packets 13 through 22?

En los frames del 13-22 se conforma la respuesta HTTP dividida en paquetes tcp, cada frame indicando que se ensamblará completamente en el 22. Por otra parte, el cliente confirma la recepción en los frame 15, 19 y 21.

j) Explain what happens in packets 23 through 26.

See the second “hint” to the left.

frame 23: se realiza otra solicitud (favicon).

frame 24: paquete tcp con parte de la respuesta HTTP que se completará en el 25.

frame 25: se completa la respuesta HTTP.

frame 26: el cliente confirma la recepción.

- k) **In one sentence describe what the user was doing (Reading email? Accessing a web page? FTP? Other?).**

El usuario está accediendo a google e imágenes e ícono se cargan automáticamente.

II) Exercise Two

Open “Wireshark”, then use the “File” menu and the “Open” command to open the file “WS-Ex- 02.pcap”. You should see 176 packets listed.

- a) **In the first few packets, the client machine is looking up the common name (cname) of a web site to find its IP address. What is the cname of this web site? Give two IP addresses for this web site.**

cname: www.yahoo.akadns.net

IPs: 216.109.117.106, 216.109.117.109

- b) **How many packets/frames does it take to receive the web page (the answer to the first http get request only)?**

16 frames

- c) **Does this web site use gzip to compress its data for sending? Does it write cookies? In order to answer these questions, look under the payload for the reassembled packet that represents the web page. This will be the last packet from question b above. Look to see if it has “Content-Encoding” set to gzip, and to see if it has a “Set-Cookie” to write a cookie.**

La respuesta usa gzip “Content-Encoding: gzip” y escribe 2 cookies (Set-Cookie).

- d) **What is happening in packets 26 and 27? Does every component of a web page have to come from the same server? See the Hint to the left.**

Parece necesitar algún dato de un servidor distinto, por lo que trata de conectarse a otro sitio.

- e) **In packet 37 we see another DNS query, this time for us.i1.yimg.com. Why does the client need to ask for this IP address? Didn't we just get this address in packet 26? (This is a trick question; carefully compare the two common names in packet 26 and 37.)**

El cname del frame 27 es us.i1.yimg.com , mientras que el del 26 es us.js2.yimg.com.

- f) **In packet 42 we see a HTTP “Get” statement, and in packet 48 a new HTTP “Get” statement. Why didn't the system need another DNS request before the second get statement? Click on packet 42 and look in the middle window. Expand the line titled “Hypertext Transfer Protocol” and read the “Host:” line. Compare that line to the “Host:” line for packet 48.**

Ambos tienen el mismo host us.i1.yimg.com, es por esto que no se necesita otro DNS.

- g) **Examine packet 139. It is one segment of a PDU that is reassembled with several other segments in packet 160. Look at packets 141, 142, and 143. Are these three packets also part of packet 160? What happens if a set of packets that are supposed to be reassembled do not arrive in a continuous stream or do not arrive in the proper order?**

El frame 143 pertenece a los que se van a terminar de ensamblar en el 160, mientras que el 141 y 142 no. Si por alguna razón los paquetes no llegan en orden o con algún error , el receptor utilizará los números de secuencia para el reordenamiento por tcp y si no llegaran, envía un ACK para la retransmisión.

- h) **Return to examine frames 141 and 142. Both of these are graphics (GIF files) from the same source IP address. How does the client know which graphic to match up to each get statement? Hint: Click on each and look in the middle window for the heading line that starts with “Transmission Control Protocol”. What difference do you see in the heading lines for the two files? Return to the original “Get” statements. Can you see the same difference in the “Get” statements?**

diferencias: Dst Port (1226 para el frame 141 y 1225 para el 142), el sequence, tamaño y ack.

similitudes: Src Port 80

Con respecto a los GET, la respuesta del frame 141 comparte el mismo Src Port del frame 133 (1226); igual que el frame 134 con la respuesta del frame 142 (1225).

III) Exercise Three

Open “Wireshark”, then use the “File” menu and the “Open” command to open the file “WS-Ex- 03.pcap”. You should see 22 packets listed.

These packets represent two different requests for web pages. Packets 1-7 involve the request for the web page www.yahoo.com. Packets 8-22 involve the request for the web page my.usf.edu.

- a) **Compare the destination port in the TCP packet in frame 3 with the destination port in the TCP packet in frame 12. What difference do you see? What does this tell you about the difference in the two requests?**

Dst Port del frame 3: 80 (request no seguro)

Dst Port del frame 12 : 443 (request seguro)

The following table compares the two requests for web pages. For example, row i) shows that frames 1-2 and frames 8-9 represent the DNS lookups for each of the web requests.

Row	www.yahoo.com frames	my.usf.com frames	Brief Explanation of Activity
i)	1-2	8-9	DNS Request to find IP address for common name & DNS Response
ii)	3-5	10-12	Three-way handshake
iii)	--	13-20	
iv)	6	21	“Get” request for web page
v)	7	22	First packet from web server with web page content.

- b) Explain what is happening in row “iii” above. Why are there no frames listed for yahoo in row “iii”?
En la fila iii) no tiene frases de Yahoo y my.usf.com sí debido a “SSL handshake with two way authentication with certificates”.
- c) **Look at the “Info” column on frame 6. It says: “GET / HTTP / 1.1. What is the corresponding Info field for the my.usf.com web request (frame 21)? Why doesn’t it read the same as in frame 6?**

El frame 6 es distinto porque está encriptado.. Se debe al handshake con SSL.

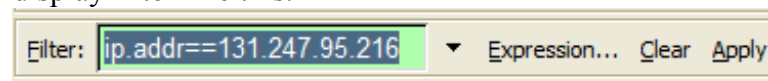
a

IV) Exercise Four

In this exercise, you are going to capture live traffic from your computer. Open up Wireshark and use the “Capture” menu to save live traffic. The Wireshark “QuickStart” guide distributed with these exercises contains more instructions on using Wireshark.

Start capturing data, visit a live web site (os.ecci.ucr.ac.cr/ci0121) using your standard Internet browser, and stop capturing data.

If you have a large amount of network traffic, the relevant data may be hidden among a lot of broadcast messages. To focus on just the key frames, you can set a display filter like this.



For the IP number enter the IP number of your client machine. Type it as shown (ip.addr==your.ip.address) in the graphic above. Then click on “Apply”.

Using an approach similar to the approach in Exercise One, describe the set of frames that you captured.

- For this description think of this as a conversation

– every discussion starts with a question and follows with an answer.

- For example, two of the frames will contain the DNS request for an IP address for the web site, and the DNS answer with the IP number.
- Remember that some answers may take several frames if they need to be reassembled from segmented packets.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.027693	192.168.40.58	17.250.98.165	UDP	83	55915 → 443 Len=41
3	0.094777	17.250.98.165	192.168.40.58	UDP	73	443 → 55915 Len=31
4	3.770030	192.168.40.58	163.178.104.62	TCP	78	58673 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2091338900 TSecr=0 SACK_PERM
5	3.893102	163.178.104.62	192.168.40.58	TCP	74	80 → 58673 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=2399823868 TSecr=2091338900 WS=128
6	3.893390	192.168.40.58	163.178.104.62	TCP	66	58673 → 80 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=2091339024 TSecr=2399823868
7	4.887882	192.168.40.58	8.8.8.8	TLSv1	250	Application Data
8	4.959528	8.8.8.8	192.168.40.58	TCP	66	443 → 58638 [ACK] Seq=1 Ack=185 Win=1010 Len=0 TSval=3043727384 TSecr=3812229052
9	4.985993	8.8.8.8	192.168.40.58	TLSv1	156	Application Data
10	4.985995	8.8.8.8	192.168.40.58	TLSv1	565	Application Data
11	4.985996	8.8.8.8	192.168.40.58	TLSv1	97	Application Data
12	4.985997	8.8.8.8	192.168.40.58	TLSv1	105	Application Data
13	4.986219	192.168.40.58	8.8.8.8	TCP	66	58638 → 443 [ACK] Seq=185 Ack=660 Win=2838 Len=0 TSval=3812229151 TSecr=3043727410
14	4.986528	192.168.40.58	8.8.8.8	TLSv1	105	Application Data
15	5.047867	8.8.8.8	192.168.40.58	TCP	66	443 → 58638 [ACK] Seq=660 Ack=224 Win=1010 Len=0 TSval=3043727473 TSecr=3812229151
18	9.405463	192.168.40.58	17.250.98.165	UDP	83	55915 → 443 Len=41
19	9.459209	17.250.98.165	192.168.40.58	UDP	74	443 → 55915 Len=32

Frame 2: Packet, 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface en0, Ethernet II, Src: e6:48:3a:87:cb:ef (e6:48:3a:87:cb:ef), Dst: SagemcomBroa_Sc:f0:f5 (b8:8c:22:00:00:00), Internet Protocol Version 4, Src: 192.168.40.58, Dst: 17.250.98.165, User Datagram Protocol, Src Port: 55915, Dst Port: 443	Data (41 bytes)	0000	b8 8c 2b 5c f0 f5 e6 48 3a 87 cb ef 00 00 45 00	...H...E...
		0010	00 45 00 00 40 00 11 dd 26 c0 a0 28 3a 11 fa	E @ 6 (:
		0020	62 a5 da 0b 01 bb 00 31 9b f5 6d 0b 63 f5 8a b3	b k 1 m c
		0030	30 b8 e4 da f6 36 41 e1 0b a3 aa 6a 9b c3 00 2e	0 6A j
		0040	14 63 27 cf 23 00 4f f2 91 97 97 20 df f2 cc 25	c 0 0 ...%
		0050	a4 98 26	.6

Pregunta: La compu quiere ingresar en el sitio os.ecci.ucr.ac.cr, pero primero necesita conectarse al servidor correspondiente.

Respuesta: La conexión inicia con un "three-way handshake" TCP entre mi IP (192.168.40.58) y el servidor (163.178.104.62):

Frame 3: SYN

Frame 4: SYN-ACK

Frame 5: ACK

Pregunta: Mi navegador envía una solicitud HTTP cifrada al servidor (a través de TLS).

Respuesta: El servidor responde también con datos cifrados. Esta comunicación se da entre los frames 6 y 11. Aquí ocurre el intercambio de certificados, claves y luego la transmisión segura del contenido.